Joint Optimization of Multiple Resources for Distributed Service Deployment in Satellite Edge Computing Networks

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***Abstract*—With the emergence of mobile edge applications and the demand for access-as-a-service, satellite mobile edge computing stands out as a disruptive technology for delivering low-latency edge service. In this article, we focus on service deployment to the edge satellites for terrestrial users, which is a key enabling technology in satellite mobile edge computing and will replace traditional centralized cloud computing. Most existing works on service deployment consider a centralized nonconvex optimization problem with high computational overhead. However, in practice, it is difficult for a single satellite to solve computationally expensive network optimization problems. To this end, we propose a distributed optimization model based on the alternating direction method of multipliers (ADMMs), which can relieve the computational burden by leveraging collaborative calculations among multiple satellites. Our proposed model minimizes the total delay of service deployment for terrestrial users by formulating a joint optimization problem that involves deployment decisions, CPU resource decisions, transmission decisions, and caching decisions. Furthermore, we propose a novel approximation method that transforms the nonconvex optimization problem to a convex one to make the joint optimization problem solvable in polynomial time. Finally, we conduct experiments using scaled global population data and show that the proposed distributed model outperforms the baselines.**

***Index Terms*—Alternating direction method of multipliers (ADMMs), distributed service deployment, multiple resources allocation, satellite edge computing network.**

I. Introduction

TERRESTRIAL networks have made significant strides in recent years, providing reliable services to a vast number of users. However, terrestrial networks face numerous challenges in specific scenarios. With the rapid spread of the Internet of Things (IoT), a large number of IoT devices, such as smart sensors and environmental monitoring devices, are in need of access network services, even in remote outdoor or maritime areas. However, the construction of ground-based communication infrastructure is extremely difficult due to high costs and harsh environmental conditions. Additionally, terrestrial networks struggle to cope with emergencies, especially natural disasters, such as earthquakes or hurricanes that severely damage communication infrastructure. These limitations prevent terrestrial networks from meeting the demands for high reliability and energy efficiency in scenarios, such as scientific expeditions and disaster relief.

To address these challenges, space-based satellite networks, such as Starlink, OneWeb, and O3b have been widely deployed. These networks are designed to meet the transmission and access needs of a diverse range of users, including those on land, at sea, and in the air. Unlike terrestrial networks, satellite networks are not hindered by natural disasters or harsh environmental conditions, making them a reliable solution for continuous connectivity in critical and remote areas.

However, traditional space-based networks are usually considered extensions of terrestrial fiber-optic networks, which aim at transmitting network information similar to a bent pipe and satisfying basic communication demands. The large propagation delay of such networks makes it difficult to satisfy the QoS requirements of emerging time-sensitive applications.

To solve this problem, satellite edge computing [1], [2], [3] which can provide low-latency services and satisfy real-time requirements based on on-board computing and edge services, has emerged and attracted great interest in both academia and industry. This grants the users access to their required services from the nearest satellite instead of from the traditional remote cloud center with the advantages of lower delay and faster response speed. Currently, many studies focusing on satellite edge computing networks have been conducted. These studies can be roughly categorized into three types based on the optimization method they use: 1) traditional centralized optimization [4], [5]; 2) the game theory [6], [7]; and 3) deep reinforcement learning [8], [9].

First, due to the limited resources and high-speed movement of satellite communication networks, the signaling and data transmission overhead is high, making it difficult for a satellite to obtain global information of the entire satellite network.

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As a result, applying traditional centralized optimization methods directly in resource-constrained satellite networks is challenging. Second, the primary idea of the game theory-based methods is to establish a game relationship between users and satellite operators by taking profits and costs into consideration. Although this type of method can usually find a Nash equilibrium under balanced interests in the established market game relationship between users and satellite network operators based on the economic principles, its focus is mostly on balancing the interests of all parties rather than achieving overall system optimization results in suboptimal algorithm performance. Third, deep reinforcement learning-based methods can leverage vast amounts of data for model training and quickly solve resource optimization problems. They can even be deployed on satellites to achieve rapid optimization results. However, these of methods are highly constrained by specific scenarios and suffer from poor generalizability, making them difficult to apply on a large scale and in diverse application domains.

Therefore, a general service deployment method that can jointly optimize multiple resources in satellite edge computing networks while meeting the delay requirement of time-sensitive services is urgently needed. The distributed algorithms proposed in existing works, such as [10] and [11] have opened new paths for solving this problem, but their models do not fully consider the characteristics of satellite edge networks. Some of them cannot adapt to the dynamic changes of the network topology; some others do not consider the multidimensional constraints of resources, such as communication, computation, and storage jointly; or treat the satellites as a single point rather than a network. To minimize the service delay in satellite edge networks, we propose a distributed service deployment framework based on the alternating direction method of multipliers (ADMMs). This framework comprehensively considers the allocation of computational, communication, and storage resources in the context of intersatellite connectivity when making service deployment decisions. The main contributions of this article are as follows.

1) We propose a distributed service deployment framework by jointly optimizing the allocation of computational, communication, and storage resources in satellite edge computing networks. This framework minimizes the total service processing delay to meet the QoS requirements of emerging time-sensitive applications in different scenarios.
2) We develop an efficient solution algorithm for the joint optimization problem based on the ADMMs. By transforming the original nonconvex problem into a convex optimization problem through relaxation and decoupling techniques, this algorithm can derive quasi-optimal service deployment decisions in polynomial time.
3) We conduct a series of simulations to evaluate the performance of the proposed service deployment framework in terms of the average task completion delay, the load density of the satellite network, and the overall performance of our proposed distribution method is very close to the result of centralized optimization. Compared with three state-of-the-art strategies, our method achieves a more balanced load distribution, lower task completion delay, and higher accuracy, with performance gains of up to 90% in satellite edge computing networks.

The remainder of this article is organized as follows. Related research on satellite edge service networks is reviewed in Section II. The system model is described in Section III. The transformation method of the original optimization problem is introduced in Section IV. In Section V, we propose a decomposition and solution method of newly derived optimization problem. Finally, the simulation results and conclusions are given in Sections VI and VII, respectively.

II. Related Work

A. Centralized Resource Optimization Methods

Centralized optimization typically relies on a single computation node using methods, such as gradient descent. Fan et al. [12] proposed an onboard service deployment policy that reduces the load on single-satellite services, thereby enhancing computational performance as evidenced by simulation results. Leyva-Mayorga et al. [9] addressed a joint optimization problem for computing and communication resources through mixed-integer programming, aiming to minimize execution delay, and demonstrated that their distributionally robust method performs comparably to state-of-art approaches. However, centralized methods face significant challenges in resource-constrained satellite networks due to high signaling overhead and difficulties in acquiring global information across satellites. This necessitates the development of a distributed satellite edge service network.

B. Game Theory-Based Resource Optimization Methods

Game theory primarily aims to model the interactions between users and satellite operators by considering their respective profits and costs, seeking to find a Nash equilibrium under balanced interests. For instance, Wang et al. [13] proposed a strategy that focuses on response time and energy consumption, demonstrating a reduction in the average cost for devices. Deng et al. [14] introduced a Stackelberg game model to design an efficient pricing mechanism that motivates service deployment. Similarly, Li et al. [15] developed a service deployment mechanism through a Stackelberg game model with both complete and incomplete information, aiming at computational resource pricing and allocation. Additionally, Galli et al. [16] proposed a potential game-based scheme for joint service deployment and resource allocation, aiming to maximize network utilization, which showed superior performance in simulations. Despite the ingenuity of the game theory in establishing a balanced market relationship, its focus on individual interests rather than overall system optimization often results in suboptimal performance.

C. Machine Learning-Based Resource Optimization Methods

Artificial intelligence learning methods utilize vast datasets for model training, enabling rapid decision making when deployed on satellites. Zhu et al. [17] presented a deep reinforcement learning algorithm to address joint service deployment and bandwidth allocation. Li et al. [18] expanded single-satellite computing to a collaborative intersatellite model, introducing a new paradigm of sensing, collaboration, and computation. Ji et al. [19] proposed a deep reinforcement learning-based solution aimed at optimizing resource utilization and increasing task completion rates. While AI methods can efficiently solve complex optimization problems by leveraging data relationships, their poor generalization across diverse scenarios limits their large-scale applicability.

Overall, many researchers are working toward deploying tasks to satellites for on-orbit processing. However, due to limitations in space computational power and communication capacity, which can easily lead to excessive overhead for global information transmission and processing, these three methods are challenging to apply directly to large-scale satellite networks. Therefore, this article adopts a distributed strategy to fully utilize the capabilities of each satellite, collaboratively optimizing to maximize network performance.

III. System Model

This section introduces the description of mathematical symbols and the network delay optimization problem, encompassing computation delay, transmission delay, propagation delay, and image pulling delay. Our objective is to minimize the total network delay.

Fig. 1 illustrates a scenario how satellite edge computing networks support user task offloading. When user devices are constrained by terminal capabilities, power consumption, or outdoor conditions and cannot handle intensive and complex computing services, satellites can assist in processing the task. Enabled by intersatellite links, nearby satellites can handover tasks to other satellites for processing if their resources are occupied. Users must determine whether to compute locally or to offload the task to satellites for processing. The workflow for deploying services in orbit involves four main steps: 1) uploading the necessary image data to the satellite; 2) transmitting user service data to the satellite; 3) performing in-orbit computing services on the satellite; and 4) sending the processed results back to the user.

TABLE I
Symbol Description

Parameter	Meaning
I_i	Image size of service i
ϕ_i	CPU cycles of service i
D_i	Data that needs to transmit for service i
E_m^{cs}	Energy consumption of satellite m
T_i^c	Node computation delay of user i
T_i^t	Network transmission delay of user i
T_i^s	Network propagation delay and cost of user i
T_i^p	Images caching delay of user i
x_{im}	Service deployment variables
c_{im}	CPU resources allocation variables
d_{im}	Caching decision variables
r_{im}	The radio of spectrum allocation variables
$\tilde{c}_{im}, \tilde{d}_{im}, \tilde{r}_{im}$	Auxiliary variables based on Eq. (11)
$\hat{x}_{im}, \hat{c}_{im}, \hat{d}_{im}$	Copy variables for distributed ADMM algorithm
$\boldsymbol{LV} = \{\widehat{\boldsymbol{x}}, \widehat{\boldsymbol{c}}, \widehat{\boldsymbol{d}}, \widehat{\boldsymbol{r}}\}$	Local variables calculated in different satellites
$\boldsymbol{GV} = \{\boldsymbol{x}, \widetilde{\boldsymbol{c}}, \widetilde{\boldsymbol{d}}\}$	Global variables
$\boldsymbol{SV} = \{\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}\}$	Scaled dual variables

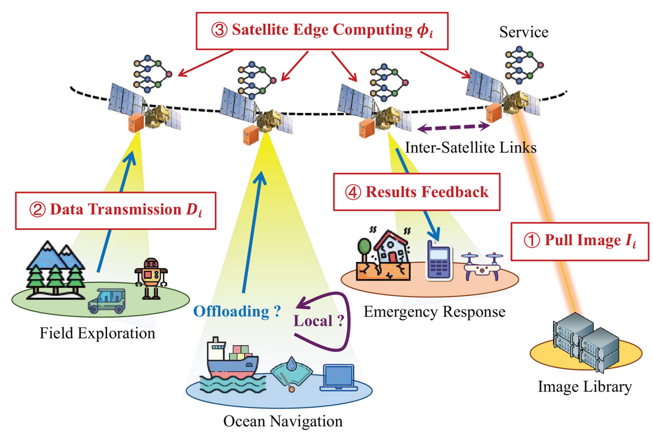

Fig. 1. Satellite edge computing networks toward different services.

A. Mathematical Description

Let $\mathcal{M} = \{1, 2, \ldots, M\}$ denote the set of satellites endowed with CPU resource and storage resource serving ground users, where satellites can communicate with each other and form a communication network. Let $\mathcal{I} = \{1, 2, \ldots, I\}$ denote the set of users that deploying services to the satellites or achieving the task by the local computing, and each user deploys one service to the satellite i.e., the number of users equals to the number of services. It is worth mentioning that local computing is more time-consuming compared with satellites offloading due to the limit of user's local computing resources. Therefore, users are willing to offload (i.e., deploy) their services to one of the satellites instead of computing locally unless offloading causes a larger delay. We assume each user has only one task that needs to deployed to satellite, and each task cannot be partitioned [20].

In addition, the three-tuple $S_i = \{I_i, \phi_i, D_i\}$ as a description of the basic information of service i is defined, where I_i denotes the size of service image i that needs to be pulled from cloud centers or update from ground users. While ϕ_i denotes the CPU cycles of service i that needs to be calculated by the users or the satellites, D_i represents the data that user i needs to be transmitted to the satellite for computation, including input parameters and data collected by users that needs to be calculated, such as some real-time data that needs to be continuously updated.

In our model, users can deploy latency-sensitive services to the satellites endowed with limited CPU resources as shown in

Fig. 1, and we use four types of decision variables to describe above process, including $\{\boldsymbol{x}, \boldsymbol{c}, \boldsymbol{d}, \boldsymbol{r}\}$, where bivariate $\boldsymbol{x} = \{x_{im}\}_{i \in \mathcal{I}, m \in \mathcal{M}}$ denotes the set of service deployment decision, that is, $x_{im} = 1$ means user i or service i deploys its service to the satellite m and $x_{im} = 0$, otherwise. While continuous variable $\boldsymbol{c} = \{c_{im}\}_{i \in \mathcal{I}, m \in \mathcal{M}}$ denotes the CPU resource that the satellites can allocate to the users who offload their services to satellites. $\boldsymbol{d} = \{d_{im}\}_{i \in \mathcal{I}, m \in \mathcal{M}}$ denotes the caching decision, which means if $d_{im} = 1$, then the satellite m cache the service i in advance, and $d_{im} = 0$ otherwise. $\boldsymbol{r} = \{r_{im}\}_{i \in \mathcal{I}, m \in \mathcal{M}}$ as a continuous variable denotes the transmission rate between satellites and ground users, and will be introduced in the next section.

B. Network Delay

1) Node Computation Delay : Let us introduce the computation delay of the service deployment model. For any user i, the computing overhead contains satellite computing or local computing, denoted as follows:

$$T_i^c = \sum_{m=1}^{M} \left(\frac{\phi_i}{c_{im}} x_{im} \right) + \frac{\phi_i}{c_i^{\text{local}}} \left(1 - \sum_{m=1}^{M} x_{im} \right) \tag{1}$$

where the first term is satellite computing delay and the second term is local computing delay. Here, c_i^{local} denotes the computational capability of user i, that is, the CPU cycles per second. To better illustrate the meaning of the above formula, let's give an example. If the user i decides to deploy its service to satellite k, then $x_{ik} = 1$ and $x_{in} = 0$ for $\forall n \in \mathcal{M}, n \neq k$ holds, and then we can conclude that $T_i^c = (\phi_i / c_{ik})$ based on (1). While if the user i decides to achieve its computation task locally, then $x_{ik} = 0$ holds for any $\forall k \in \mathcal{M}$, meaning that $\sum_{m=1}^{M} x_{im} = 0$, and then we can conclude that $T_i^c = (\phi_i / c_i^{\text{local}})$ based on (1).

In addition, considering that the energy of each satellite is limited, thus excessive task offloading will exceed satellite burden capacity. Based on the above considerations, let us give the formula for the energy consumption of each satellite as follows when users offload their services to satellite:

$$E_m^{cs} = \sum_{i=1}^{I} E_{im}^{cs} = \sum_{i=1}^{I} \left[\varepsilon_2 (c_{im})^2 x_{im} \right] \tag{2}$$

where E_{im}^{cs} denotes the energy consumption [21] of user i if the user offloads its computation task to satellite m with the CPU resource allocation of c_{im}. Besides, ε_2 denotes the factor depending on the chip architecture proposed in [10] and [22].

2) Network Transmission Delay : Each user in the satellite network is linked to one of the satellites with the case of the seamless global coverage. Therefore, the user i can upload the input parameters as well as the real-time data to a certain satellite with the size of D_i when $\sum_{m=1}^{M} x_{im} \neq 0$, and the transmission rate can be allocated between the terrestrial users and satellites. Without loss of generality, we consider the case when spectrum is overlaid in the satellite network, i.e., there is spectrum interference among the ground users. According to the Shannon formula, the spectrum efficiency of user i with respect to the access satellite m is listed as follows:

$$e_{im} = \log_2 \left[1 + \frac{g_{im} p_i}{\sum_{j \in \mathcal{I}_m, j \neq i} g_{jm} p_j + \sigma^2} \right] \tag{3}$$

where $\mathcal{I}_m$ as a subset of $\mathcal{I}$ means the set of users that covered by the access satellite m and the existence of spectrum interference with other users. Besides, p_i is the transmission power of user i, and g_{im} is the channel gain between user i and access satellite m.

Let us define $r_{im} \in [0, 1]$ as the radio spectrum allocation profile to user i for the access satellite m, and the transmission delay for user i can be calculated as follows:

$$T_i^t = \sum_{m=1}^{M} s_{im} T_{im}^t = \sum_{m=1}^{M} \left[s_{im} \left(\frac{\sum_{u=1}^{M} x_{iu} D_i}{r_{im} B e_{im}} \right) \right] \tag{4}$$

where $s_{im} = 1$ as a binary variable denotes user i covered by satellite m (called access satellite for simplicity), and $s_{im} = 0$ otherwise. s_{im} is known and given, which can be seen as a constant. In addition, B is the bandwidth in between satellite and users, and Be_{im} is the total transmission rate based on the Shannon formula. While r_{im} is the ratio of Be_{im} [11], which can be divided by the satellite system. We assume that each user is covered by only one satellite, which means $\sum_{m=1}^{M} s_{im} = 1$ for all users, and the transmission delay between user i and access satellite m is T_{im}^t. Note that if satellite m does not cover user i, then there is no need to calculate T_{im}^t since $s_{im} = 0$. Besides, if user i within the coverage area of satellite m does not offload its computation task to any satellite, then T_{im}^t equals to 0 since $\sum_{u=1}^{M} x_{iu} = 0$. Note that $s_{im} = 0$ does not means $x_{im} = 0$.

3) Network Propagation Delay and Cost : Considering that each satellite is endowed with limited computation resource and storage resource cannot satisfy the demands of computation-intensive offloading requests due to the nonuniform distribution of users on the earth illustrated as Fig. 2. Users may offload their offloading requests to other satellites causing propagation delay among the satellites, and the formula of propagation delay is given by

$$\begin{aligned} T_i^s &= T_i^{\text{Propagation}} + T_i^{\text{Penalty}} \\ &= \sum_{m=1}^{M} \left[\frac{\text{DIS}_{im}}{c} x_{im} \right] + \sum_{m=1}^{M} [J_{im} k_1 (1 - s_{im}) x_{im}] \end{aligned} \tag{5}$$

where J_{im} and DIS_{im} denote the routing hops and the distance between the satellite m running computation task for user i and the access satellite covering user i, respectively. Here, the first term $T_i^{\text{Propagation}}$ means propagation delay between users and satellites, where c denotes the speed of the light. It is evident that if user i offloads its computation task to satellite k, then only x_{ik} equals to 1, and the first term can be simplified to $T_i^{\text{Propagation}} = \text{DIS}_{ik}/c$. Considering that satellite resources are precious, we expect each user to offload its computation task to the access satellite which can help avoid extra intersatellite propagation burden. The second term T_i^{Penalty} means the cost of punishment if the user do not offload their computation task to the access satellite (i.e., $s_{im} = 0$ and $x_{im} = 1$), and k_1 is the penalty coefficient.

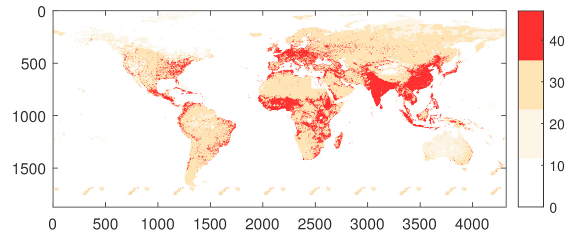

Fig. 2. Distribution of ground users according to population density and the brighter the color, the more the population. A nonlinear transformation of the population data is applied for the convenience of presentation.

4) Images Caching Delay : Define bivariate d_{im} as the caching decision of the satellites, which means if $d_{im} = 1$ then satellite m caches the computation task of user i with the data size of I_i and $d_{im} = 0$ otherwise, where the data size I_i is also called service images, including source codes and dependencies. Based on the above description, the delay for pulling images can be calculated as

$$T_i^p = \sum_{m=1}^{M} (1 - d_{im}) x_{im} (k_2 I_i) \tag{6}$$

where coefficient k_2 represents the effect of the service images on the images pulling delay. It is worth noting that (6) contains x_{im}, which means if user i does not offload its computation task to the satellite m, then $T_i^p = 0$. In addition, if satellite m caches the computation task (i.e., service image) of user i, then $T_i^p = 0$ due to $d_{im} = 1$. There is only one case that T_i^p is not equals to 0, that is, user i offloads its computation task to the satellite m but satellite m does not cache that.

C. Problem Formulation

Now, let us introduce the objective function as well as the optimization problem for the satellite edge computing. First, our objective function is defined as total network delay for all users, including computation delay, transmission delay, propagation delay, as well as image pulling delay

$$\begin{aligned}
&T_1^{\text{total}}(\boldsymbol{x}, \boldsymbol{c}, \boldsymbol{d}, \boldsymbol{r}) \\
&= \sum_{i=1}^{I} \left(T_i^c + 2T_i^s + T_i^p + T_i^t\right) \\
&= \sum_{m=1}^{M} \sum_{i=1}^{I} \underbrace{\frac{s_{im} \sum_{u=1}^{M} x_{iu} D_i}{r_{im} B \log_2\left(1 + \frac{g_{im} p_i}{\sum_{j \in \mathcal{I}_m^k, j \neq i} g_{jm} p_j + \sigma^2}\right)}}_{\text{local}} \\
&\quad + \sum_{m=1}^{M} \sum_{i=1}^{I} \underbrace{\left[-\frac{\phi_i}{c_i^{\text{local}}} + \frac{2\text{DIS}_{im}}{c} + k_2 I_i + J_{im} k_1 (1 - s_{im})\right] x_{im}}_{\text{global}} \\
&\quad + \sum_{m=1}^{M} \sum_{i=1}^{I} \underbrace{\left(\frac{\phi_i}{c_{im}}\right) x_{im}}_{\text{global}} + \sum_{m=1}^{M} \sum_{i=1}^{I} \underbrace{(-d_{im} k_2 I_i) x_{im}}_{\text{global}} + \sum_{i=1}^{I} \underbrace{\frac{\phi_i}{c_i^{\text{local}}}}_{\text{constant}}
\end{aligned} \tag{7}$$

$$= \sum_{m=1}^{M} \sum_{i=1}^{I} \left[h(x_{im}, r_{im}) + g(x_{im}) + f(x_{im}, c_{im}) + q(x_{im}, d_{im})\right] + c_0. \tag{8}$$

For concision, we use $h(x_{im}, r_{im})$, $g(x_{im})$, $f(x_{im}, c_{im})$, $q(x_{im}, d_{im})$, and constant c_0 rewrite the total network delay $T_1^{\text{total}}(\boldsymbol{x}, \boldsymbol{c}, \boldsymbol{d}, \boldsymbol{r})$, where $h(x_{im}, r_{im})$ is short for the first term of (7) demonstrating transmission delay, and $g(x_{im})$ is short for the second term of (7) with respect to variable x_{im}. Here, the third term is denoted as $f(x_{im}, c_{im})$ which is short for computation delay, the forth term is represented as $q(x_{im}, d_{im})$ demonstrating image pulling delay, and the final term c_0 is constant which is not related to decision variables.

Based on the above description, we formulate the total network delay minimization problem as follows:

$$\boldsymbol{P}_1\textbf{:} \min_{\boldsymbol{x}, \boldsymbol{c}, \boldsymbol{d}, \boldsymbol{r}} T_1^{\text{total}}(\boldsymbol{x}, \boldsymbol{c}, \boldsymbol{d}, \boldsymbol{r}) \tag{9}$$

$$\text{s.t.} \begin{cases}
\sum_{m=1}^{M} x_{im} \leq 1 \quad \forall i \\
\sum_{i=1}^{I} x_{im} c_{im} \leq C_0 \quad \forall m \\
\sum_{i=1}^{I} x_{im} d_{im} \leq D_0 \quad \forall m \\
\sum_{i=1}^{I} s_{im} \left(\sum_{u=1}^{M} x_{iu}\right) r_{im} \leq R_0 \quad \forall m \\
\sum_{i=1}^{I} \left[\varepsilon_2 (c_{im})^2 x_{im}\right] \leq E_{\text{sate}} \quad \forall m \\
x_{im}, d_{im} \in \{0, 1\}, c_{im} \geq 0, 0 \leq r_{im} \leq 1 \ \forall m, i
\end{cases} \tag{10}$$

where C_0, R_0, D_0, and E_{sate} represents the capacity of the satellite about computing capacity, transmission capacity, cache capability, and energy upper, respectively. While the first constraint is the offloading constraint, demonstrating that each user can offload its computation task to at most one satellite. and the second constraint and the third constraint represent the upper limit constraint of each satellite CPU resource capacity and caching resource capacity, respectively. While the forth denotes the limit of transmission capacity for each access satellite, and the fifth constraint makes the energy consumption of each satellite not exceed the upper limit.

IV. Problem Convexification and Variables Copy

In order to solve the optimization problem P_1 and minimize the objection function of the satellite network delay, two steps are proposed, including problem transformation and variables copy. Specifically, problem transformation aims at transforming the optimization problem P_1 into the convex optimization problem P_2 by variable substitution and binary variable relaxation. Besides, variables copy aims at making the local copy for each satellite, which can make each satellite participate in the decision making based on local variables in the next decomposition section.

A. Problem Transformation

The optimization problem P_1 is a mixed integer optimization problem, which is known as an NP-hard problem, that is difficult to solve. Thus, we propose an approximate convex optimization method, including variable substitution and binary variable relaxation. First, let us replace the decision variables $\{c_{im}\}_{i \in \mathcal{I}, m \in \mathcal{M}}$, $\{d_{im}\}_{i \in \mathcal{I}, m \in \mathcal{M}}$ and $\{r_{im}\}_{i \in \mathcal{I}, m \in \mathcal{M}}$ with the new decision variables $\{\tilde{c}_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}$, $\{\tilde{d}_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}$ and $\{\tilde{r}_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}$ by the following formula, respectively:

$$\begin{cases} \tilde{c}_{im} = x_{im}c_{im} \\ \tilde{d}_{im} = x_{im}d_{im} \\ \tilde{r}_{im} = \left(\sum_{u=1}^{M} x_{iu}\right) r_{im}. \end{cases} \tag{11}$$

Next, the objective function, including $h(x_{im}, \tilde{r}_{im})$, $f(x_{im}, \tilde{c}_{im})$, and $q(\tilde{d}_{im})$ can be equivalently transformed or approximately transformed into the following equation based on the variable substitution proposed in (11):

$$\begin{cases} h(x_{im}, \tilde{r}_{im}) = \frac{s_{im}Mx_{iu}^2 D_i}{\tilde{r}_{im}Be_{im}} \\ f(x_{im}, \tilde{c}_{im}) = \left(\frac{\phi_i}{\tilde{c}_{im}}\right)x_{im}^2 \\ q\left(\tilde{d}_{im}\right) = -\tilde{d}_{im}k_2 I_i. \end{cases} \tag{12}$$

Considering that if a user chooses to offload its service to a satellite instead of computing locally, then the user can only select at most one satellite to offload its service, that is, there is at most one nonzero x_{im} in vector $\{x_{i1}, x_{i2}, \ldots, x_{iM}\}$. Therefore, $x_{im_1}x_{im_2} = 0$ always holds when $x_{im_1}, x_{im_2} \in \{x_{i1}, x_{i2}, \ldots, x_{iM}\}$ and m_1 not equals to m_2 for all i. Based on such conclusion, the following formula holds $(\sum_{u=1}^{M} x_{iu})^2 = \sum_{u=1}^{M} x_{iu}^2$, and $h(x_{im}, \tilde{r}_{im})$ can be simplified to $([s_{im}\sum_{u=1}^{M} x_{iu}^2 D_i]/[\tilde{r}_{im}Be_{im}])$. Since the above term needs to sum over m and i, the summation term for u is repeated and can be simplified as $([s_{im}Mx_{iu}^2 D_i]/[\tilde{r}_{im}Be_{im}])$. Then, based on the variable substitution proposed in (11), the constraints in the optimization problem P_1 can be transformed, yielding the following new optimization problem:

$$\boldsymbol{P_2}\text{:} \min_{\boldsymbol{x},\tilde{\boldsymbol{c}},\tilde{\boldsymbol{d}},\tilde{\boldsymbol{r}}} T_2^{\text{total}}\left(\boldsymbol{x}, \tilde{\boldsymbol{c}}, \tilde{\boldsymbol{d}}, \tilde{\boldsymbol{r}}\right) \tag{13}$$

$$\text{s.t.} \begin{cases} \sum_{m=1}^{M} x_{im} \le 1 \quad \forall i \\ \sum_{i=1}^{I} \tilde{c}_{im} \le C_0 \quad \forall m \\ \sum_{i=1}^{I} \tilde{d}_{im} \le D_0 \quad \forall m \\ \sum_{i=1}^{I} s_{im}\tilde{r}_{im} \le R_0 \quad \forall m \\ \sum_{i=1}^{I} \frac{\varepsilon_2 \tilde{c}_{im}^2}{x_{im}} \le E_{\text{sate}} \quad \forall m \\ x_{im}, \tilde{d}_{im} \in \{0,1\}, \tilde{c}_{im} \ge 0, 0 \le \tilde{r}_{im} \le 1 \quad \forall m, i \\ x_{im} \ge \frac{\tilde{c}_{im}}{C_0}, x_{im} \ge \tilde{d}_{im}, \sum_{u=1}^{M} x_{iu} \ge \tilde{r}_{im} \quad \forall m, i. \end{cases} \tag{14}$$

We add three constraints, including $x_{im} \ge (\tilde{c}_{im}/C_0), x_{im} \ge \tilde{d}_{im}$ and $\sum_{u=1}^{M} x_{iu} \ge \tilde{r}_{im}$, which are derived from (11) using the value range of $c_{im} \le C_0$, $d_{im} \le 1$, and $r_{im} \le 1$ for all i, m. It turns out that the optimization problem P_2 is equivalent to the optimization problem P_1 based on variable substitution. Therefore, we can solve the optimization problem P_2 instead of P_1, followed by recovering the variables c_{im}, d_{im}, r_{im} through (11). However, in the process of recovering the variables from $\tilde{c}_{im}$ to c_{im}, if x_{im} equals to 0, then (11) always holds regardless of the value of c_{im}. To solve such problems, the following recovering method is proposed according to the meaning of symbolic meaning x_{im}, i.e.,:

$$c_{im} = \begin{cases} \frac{\tilde{c}_{im}}{x_{im}}, & x_{im} > 0 \\ 0, & x_{im} = 0 \end{cases} \tag{15}$$

$$d_{im} = \begin{cases} \frac{\tilde{d}_{im}}{x_{im}}, & x_{im} > 0 \\ 0, & x_{im} = 0 \end{cases} \tag{16}$$

$$r_{im} = \begin{cases} \frac{\tilde{r}_{im}}{\sum_{u=1}^{M} x_{iu}}, & \sum_{u=1}^{M} x_{iu} > 0 \\ 0, & \sum_{u=1}^{M} x_{iu} = 0. \end{cases} \tag{17}$$

Specifically, considering that if the user i does not offload service to the satellite m ($x_{im} = 0$), then the satellite m does not need to allocate resources to user i ($c_{im} = 0$) as (15) shown. Therefore, we define $c_{im} = 0$ when $x_{im} = 0$ to recover the above variable substitution. Likewise, the same definitions and substitutions apply to d_{im} and r_{im}.

B. Convexification

After transforming the optimization problem P_1 into the optimization problem P_2 by variable substitution, it is evident that P_2 is still a nonconvex problem due to the binary variables $\boldsymbol{x} = \{x_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}$ and $\boldsymbol{d} = \{d_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}$, which is difficult to solve. Thus, we relax each $x_{im} \in \{0,1\}$ and $d_{im} \in \{0,1\}$ into continuous variable, which means $0 \le x_{im} \le 1$ and $0 \le d_{im} \le 1$ holds for all $i \in \mathcal{I}, m \in \mathcal{M}$, and the continuous variable x_{im}, d_{im} can be understood as the preference of offloading decision and the preference of caching decision for user i. Based on the above transformation, a new convex optimization problem P_3 is proposed as follows:

$$\boldsymbol{P_3}\text{:} \min_{\boldsymbol{x},\tilde{\boldsymbol{c}},\tilde{\boldsymbol{d}},\tilde{\boldsymbol{r}}} T_3^{\text{total}}\left(\boldsymbol{x}, \tilde{\boldsymbol{c}}, \tilde{\boldsymbol{d}}, \tilde{\boldsymbol{r}}\right) \tag{18}$$

$$\text{s.t.} \begin{cases} \sum_{m=1}^{M} x_{im} \le 1 \quad \forall i \\ \sum_{i=1}^{I} \tilde{c}_{im} \le C_0 \quad \forall m \\ \sum_{i=1}^{I} \tilde{d}_{im} \le D_0 \quad \forall m \\ \sum_{i=1}^{I} s_{im}\tilde{r}_{im} \le R_0 \quad \forall m \\ \sum_{i=1}^{I} \frac{\varepsilon_2 \tilde{c}_{im}^2}{x_{im}} \le E_{\text{sate}} \quad \forall m \\ 0 \le x_{im}, \tilde{d}_{im}, \tilde{r}_{im} \le 1, \quad \tilde{c}_{im} \ge 0 \quad \forall m, i \\ x_{im} \ge \frac{\tilde{c}_{im}}{C_0}, x_{im} \ge \tilde{d}_{im}, \sum_{u=1}^{M} x_{iu} \ge \tilde{r}_{im} \quad \forall m, i \end{cases} \tag{19}$$

where the objective function $T_3^{\text{total}}(\boldsymbol{x}, \tilde{\boldsymbol{c}}, \tilde{\boldsymbol{d}}, \tilde{\boldsymbol{r}})$, including $h(x_{im}, \tilde{r}_{im})$, $g(x_{im})$, $q(\tilde{d}_{im})$, and $f(x_{im}, \tilde{c}_{im})$ is denoted as follows:

$$\begin{aligned} &T_3^{\text{total}}\left(\boldsymbol{x}, \tilde{\boldsymbol{c}}, \tilde{\boldsymbol{d}}, \tilde{\boldsymbol{r}}\right) \\ &= \sum_{m=1}^{M}\sum_{i=1}^{I}\Big[h(x_{im}, \tilde{r}_{im}) + g(x_{im}) + q\left(\tilde{d}_{im}\right) \\ &\quad + f(x_{im}, \tilde{c}_{im})\Big] + c_0. \end{aligned} \tag{20}$$

Considering that c_0 is a constant, we thus ignore this part in the following derivation.

Proposition 1: If the optimization problem P_3 is feasible, it is a jointly convex optimization problem with respect to all optimization variables $\boldsymbol{x}, \boldsymbol{c}, \boldsymbol{d}$, and $\boldsymbol{r}$.

Proof: The convexity of the optimization problem P_3 is proved based on *quadratic-over-linear function*, and the similar proof can be found in [4]. For the sake of simplicity, let vector χ denote all of the decision variables, including $x_{im}, \tilde{c}_{im}, \tilde{d}_{im}$, and $\tilde{r}_{im}$, that is, $\chi = \{x_{11}, \ldots, x_{IM}, \tilde{c}_{11}, \ldots, \tilde{c}_{IM}, \tilde{d}_{11}, \ldots, \tilde{d}_{IM}, \tilde{r}_{11}, \ldots, \tilde{r}_{IM}\}$. Then, let's consider the convexity of the *quadratic-over-linear function* with the expression of $\mathcal{W}(\chi_1, \chi_2) = (\chi_1^2/\chi_2)$ for any

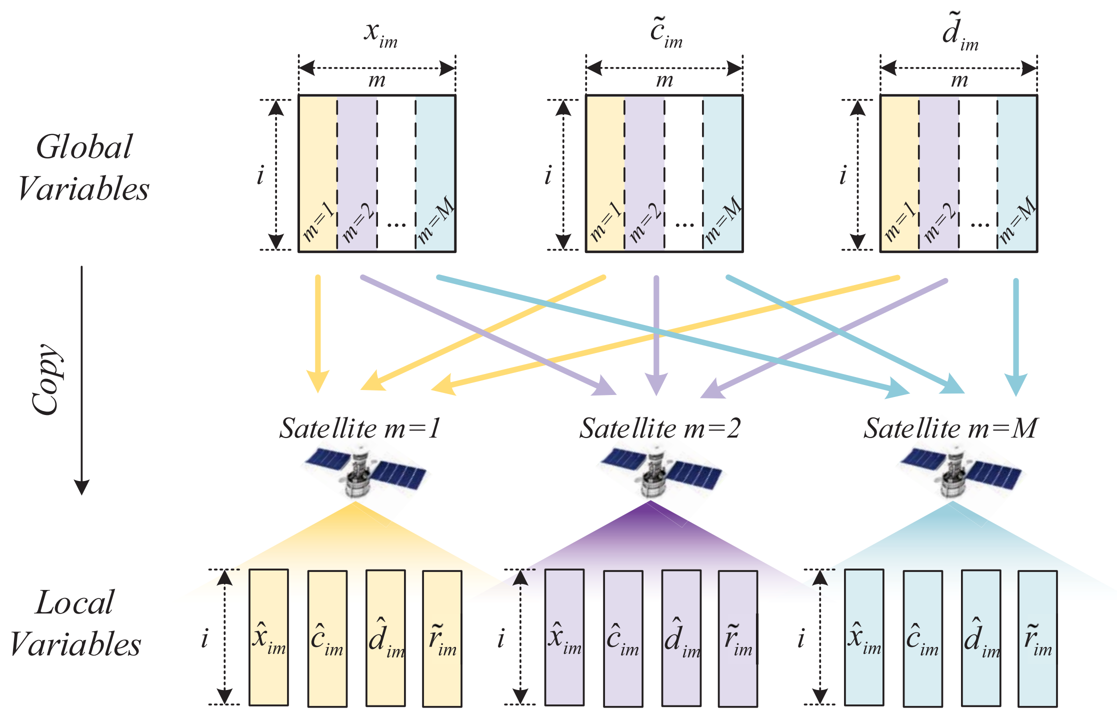

Fig. 3. Each satellite m stores its own local variables, i.e., vector $\{r_{im}\}_{i\in\mathcal{I}}$, as well as the copy of global variables, including vector $\widehat{\boldsymbol{x}}_{\boldsymbol{m}} = \{\hat{x}_{im}\}_{i\in\mathcal{I}}$, $\widehat{\boldsymbol{c}}_{\boldsymbol{m}} = \{\hat{c}^m_{in}\}_{i\in\mathcal{I}}$, and $\widehat{\boldsymbol{d}}_{\boldsymbol{m}} = \{\hat{d}_{im}\}_{i\in\mathcal{I}}$.

$\chi_1 \in \chi$ and $\chi_2 \in \chi$. Then, the Hessian matrix of the *quadratic-over-linear function* $\mathcal{W}(\chi)$ can be calculated as follows:

$$\begin{aligned}\nabla^2\mathcal{W}(\chi) &= \frac{2}{\chi_2^3}\left[2\,\boldsymbol{diag}\left(\chi_2^2, \chi_1^2\right) - \mathcal{Z}\mathcal{Z}^T\right]\\ &= \frac{2}{\chi_2^3}\begin{bmatrix}\chi_2\\ -\chi_1\end{bmatrix}\begin{bmatrix}\chi_2\\ -\chi_1\end{bmatrix}^T \succeq 0\end{aligned} \tag{21}$$

where vector $\mathcal{Z} = [\chi_2, \chi_1]^T$ is defined in order to simplify the notation. It is evident that the matrix $\nabla^2\mathcal{W}(\chi)$ is the positive semi-definite based on (21), which means the quadratic-over-linear function $\mathcal{W}(\chi)$ is convex for any $\chi_1 \in \chi$ and $\chi_2 \in \chi$ according to the second-order conditions.

Besides, considering that *non-negative weighted sums* is the operation that preserves convexity of functions, meaning that if $\mathcal{F}_i$ is a convex function for all i, then $\sum_i \alpha_i\mathcal{F}_i$ is also a convex function if $\alpha_i > 0$. Considering that $q(\tilde{d}_{im})$ and $g(x_{im})$ are the linear functions with respect to variables $\tilde{d}_{im}$ and x_{im}, respectively, $f(x_{im}, \tilde{c}_{im})$ is the quadratic-over-linear function of variables x_{im} and $\tilde{c}_{im}$, and $h(x_{im}, \tilde{r}_{im})$ is the sum of quadratic-over-linear function with respect to variables x_{im} and $\tilde{r}_{im}$, all of the above functions are convex functions with non-negative weighted sums according to (20). Thus, following the preceding principle, the objective function $T_3^{\text{total}}(\boldsymbol{x}, \tilde{\boldsymbol{c}}, \tilde{\boldsymbol{d}}, \tilde{\boldsymbol{r}})$ of the optimization problem P_3 is a convex function.

After analyzing the objective function, let us focus on the constraints in the optimization problem P_3. It is evident that the constraints are all linear constraints or convex constraints (i.e., non-negative weighted sums function). Therefore, it is a jointly convex optimization problem with respect to all optimization variables $\boldsymbol{x}, \boldsymbol{c}, \boldsymbol{d}$, and $\boldsymbol{r}$. Note that, the transformation from the optimization problems P_1 to P_3 in this article is an approximate transformation especially for the objective function, for which we will show its algorithm advantage. ■

C. Copy of Global Variables

There are two types of variables in our paper, one is local variables, i.e., $\tilde{\boldsymbol{r}} = \{\tilde{r}_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}$, the other is global variables, i.e., $\boldsymbol{x} = \{x_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}$, $\tilde{\boldsymbol{c}} = \{\tilde{c}_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}$, $\tilde{\boldsymbol{d}} = \{\tilde{d}_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}$. The difference between local variables and global variables is that they belong to different scopes, that is, the local variables exist only in one satellite, while global variables exist in all satellites. Since global variables are related to all satellites, each global variable will create corresponding copy for each satellite, and each copy is bundled to only one satellite. In this way, each satellite can participate in the decision-making and optimize the copy of global variables based on user requests. Thus, the copies of global variables are also called local variables, which can be independently optimized by each satellite locally.

Specifically, following the previous concept introduction about local variables and global variables, we create the copies of global variables $\boldsymbol{x} = \{x_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}$, $\tilde{\boldsymbol{c}} = \{\tilde{c}_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}$, $\tilde{\boldsymbol{d}} = \{\tilde{d}_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}$ for each satellite m, and denote them as $\widehat{\boldsymbol{x}}$, $\widehat{\boldsymbol{c}}$, and $\widehat{\boldsymbol{d}}$:

$$\begin{cases}\widehat{\boldsymbol{x}} = \{\widehat{\boldsymbol{x}}_{\boldsymbol{m}}\}_{m\in\mathcal{M}} = \{\hat{x}_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}\\ \widehat{\boldsymbol{c}} = \{\widehat{\boldsymbol{c}}_{\boldsymbol{m}}\}_{m\in\mathcal{M}} = \{\hat{c}_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}\\ \widehat{\boldsymbol{d}} = \{\widehat{\boldsymbol{d}}_{\boldsymbol{m}}\}_{m\in\mathcal{M}} = \{\hat{d}_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}\end{cases} \tag{22}$$

where symbol m is used to denote the index of satellites and symbol i is used to denote the index of service. And the relationship among the multiple variables is illustrated in Fig. 3.

However, different satellites optimize the copies of global variables independently, causing inconsistencies with the copies that are duplicated from the same global variables. In order to keep local variables consistent with each other, the following consensus [23], [24], [25] formula must hold:

$$\begin{cases} \hat{x}_{im} = x_{im} & \forall m, i \\ \hat{c}_{im} = \tilde{c}_{im} & \forall m, i \\ \hat{d}_{im} = \tilde{d}_{im} & \forall m, i. \end{cases} \tag{23}$$

Considering that $\tilde{r}_{im}$ is a local variable, thus we do not need to rebuild it from the global variables. Next, the object function can be rewritten based on the local variables as follows:

$$\begin{aligned} &T_4^{\text{total}}\left(\hat{\boldsymbol{x}}, \hat{\boldsymbol{c}}, \hat{\boldsymbol{d}}, \tilde{\boldsymbol{r}}\right) \\ &= \sum_{m=1}^{M}\sum_{i=1}^{I}\Big[h\left(\hat{x}_{im}, \tilde{r}_{im}\right) + g\left(\hat{x}_{im}\right) + q\left(\hat{d}_{im}\right) \\ &\quad + f\left(\hat{x}_{im}, \hat{c}_{im}\right)\Big] \end{aligned} \tag{24}$$

where the objective function, including $h(\hat{x}_{im}, \tilde{r}_{im})$, $g(\hat{x}_{im})$, $q(\hat{d}_{im})$, and $f(\hat{x}_{im}, \hat{c}_{im})$ can be rewritten as follows based on the local variables:

$$\begin{cases} h\left(\hat{x}_{im}, \tilde{r}_{im}\right) = \frac{s_{im} M \hat{x}_{iu}^2 D_i}{\tilde{r}_{im} B e_{im}} \\ g\left(\hat{x}_{im}\right) = \left[-\frac{\phi_i}{c_i^{\text{local}}} + \frac{2\text{DIS}_{im}}{c} + k_2 I_i + J_{im} k_1 (1 - s_{im})\right]\hat{x}_{im} \\ q\left(\hat{d}_{im}\right) = -\hat{d}_{im} k_2 I_i \\ f\left(\hat{x}_{im}, \hat{c}_{im}\right) = \left(\frac{\phi_i}{\hat{c}_{im}}\right)\hat{x}_{im}^2. \end{cases} \tag{25}$$

Next, the equivalent consensus optimization problem can be rewritten as P_4, including both local variables and global variables

$$\boldsymbol{P_4}\text{:} \min_{\substack{\{\hat{\boldsymbol{x}}, \hat{\boldsymbol{c}}, \hat{\boldsymbol{d}}, \tilde{\boldsymbol{r}}\} \\ \{\boldsymbol{x}, \tilde{\boldsymbol{c}}, \tilde{\boldsymbol{d}}\}}} T_4^{\text{total}}\left(\hat{\boldsymbol{x}}, \hat{\boldsymbol{c}}, \hat{\boldsymbol{d}}, \tilde{\boldsymbol{r}}\right) \tag{26}$$

$$\text{s.t.} \begin{cases} \sum_{m=1}^{M} x_{im} \le 1 \quad \forall i \\ \sum_{i=1}^{I} \hat{c}_{im} \le C_0 \quad \forall m \\ \sum_{i=1}^{I} \hat{d}_{im} \le D_0 \quad \forall m \\ \sum_{i=1}^{I} s_{im} \tilde{r}_{im} \le R_0 \quad \forall m \\ \sum_{i=1}^{I} \frac{\varepsilon_2 \hat{c}_{im}^2}{\hat{x}_{im}} \le E_{\text{sate}} \quad \forall m \\ 0 \le \hat{x}_{im}, \hat{d}_{im}, \tilde{r}_{im} \le 1, \quad \hat{c}_{im} \ge 0 \quad \forall m, i \\ \hat{x}_{im} \ge \frac{\hat{c}_{im}}{C_0}, \hat{x}_{im} \ge \hat{d}_{im}, \sum_{u=1}^{M} \hat{x}_{iu} \ge \tilde{r}_{im} \quad \forall m, i \\ \hat{x}_{im} = x_{im}, \hat{c}_{im} = \tilde{c}_{im}, \hat{d}_{im} = \tilde{d}_{im} \quad \forall m, i. \end{cases} \tag{27}$$

The above constraints can be divided into two categories, including global constraints and local constraints, where the first constraints are the global constraints for i, while other constraints are the local constraints from the second constraints to the final constraints for m. The reason for defining the first constraints as global constraints is that each x_{im} is added across multiple satellites, which cannot be effectively limited and achieved by only one satellite.

Next, we will derive the subproblems according to the optimization problem P_4. For ease of presentation, we define the feasible set of local variables for each satellite m as follows [10] and [11]:

$$\phi_m = \left\{ \begin{array}{c|c} \begin{array}{c} \hat{\boldsymbol{x}}_{\boldsymbol{m}} \\ \hat{\boldsymbol{c}}_{\boldsymbol{m}} \\ \hat{\boldsymbol{d}}_{\boldsymbol{m}} \\ \tilde{\boldsymbol{r}}_{\boldsymbol{m}} \end{array} & \begin{array}{c} \sum_{i=1}^{I} \hat{c}_{im} \le C_0 \\ \sum_{i=1}^{I} \hat{d}_{im} \le D_0 \\ \sum_{i=1}^{I} s_{im} \tilde{r}_{im} \le R_0 \\ \sum_{i=1}^{I} \frac{\varepsilon_2 \hat{c}_{in}^2}{\hat{x}_{im}} \le E_{\text{sate}} \\ 0 \le \hat{x}_{im}, \hat{d}_{im}, \tilde{r}_{im} \le 1, \ \hat{c}_{im} \ge 0 \quad \forall i \\ \hat{x}_{im} \ge \frac{\hat{c}_{im}}{C_0}, \hat{x}_{im} \ge \hat{d}_{im}, \sum_{u=1}^{M} \hat{x}_{iu} \ge \tilde{r}_{im} \quad \forall i \end{array} \end{array} \right\} \forall m.$$

Then, optimization problem P_4 can be equivalently replaced with the optimization problem P_5 illustrated as follows:

$$\boldsymbol{P_5}\text{:} \min_{\substack{\{\hat{\boldsymbol{x}}, \hat{\boldsymbol{c}}, \hat{\boldsymbol{d}}, \tilde{\boldsymbol{r}}\} \\ \{\boldsymbol{x}, \tilde{\boldsymbol{c}}, \tilde{\boldsymbol{d}}\}}} \sum_{m=1}^{M} y_m\left(\hat{x}_{im}, \hat{c}_{im}, \hat{d}_{im}, \tilde{r}_{im}\right) \tag{28}$$

$$\text{s.t.} \begin{cases} \hat{x}_{im} = x_{im} & \forall m, i \\ \hat{c}_{im} = \tilde{c}_{im} & \forall m, i \\ \hat{d}_{im} = \tilde{d}_{im} & \forall m, i \end{cases} \tag{29}$$

where the feasible set for local variables is $\{\hat{x}_{im}, \hat{c}_{im}, \hat{d}_{im}, \tilde{r}_{im}\}_{i \in \mathcal{I}, m \in \mathcal{M}} \in \phi_m$. The objective function is the sum of each satellite, where each satellite focuses on optimizing its own objective function $y_m(\hat{x}_{im}, \hat{c}_{im}, \hat{d}_{im}, \tilde{r}_{im})$ that is shown as follows:

$$\begin{aligned} &y_m\left(\hat{x}_{im}, \hat{c}_{im}, \hat{d}_{im}, \tilde{r}_{im}\right) \\ &= \begin{cases} \sum_{i=1}^{I}\Big[h\left(\hat{x}_{im}, \tilde{r}_{im}\right) + g\left(\hat{x}_{im}\right) + q\left(\hat{d}_{im}\right) + f\left(\hat{x}_{im}, \hat{c}_{im}\right)\Big] \\ \qquad \text{if } \{\hat{x}_{im}, \hat{c}_{im}, \hat{d}_{im}, \tilde{r}_{im}\}_{i \in \mathcal{I}, n \in \mathcal{M}} \in \phi_m \\ +\infty, \quad \text{otherwise.} \end{cases} \end{aligned} \tag{30}$$

Note that, the objective function, the constraint, and the feasible domain in the optimization problem P_5 are separable with respect to variable m, which can be decomposed in the next section by ADMM [26], [27], [28].

V. Problem Decomposition and Solving

In this section, the above convex optimization problem P_5 is decomposed into multiple distributed subproblems by ADMM, which aims at relieving the computing burden of a single satellite node, and making each satellite participate in the decision making. By this decomposition method, each satellite can independently optimize the local variables and the copy of global variables. We shall introduce our distributed decomposition and solving method, including the problem decomposition, followed by presenting the variables update algorithm as well as a binary recovery algorithm.

A. Problem Decomposition

First, the augmented Lagrangian of P_5 can be written as follows:

$$\begin{aligned} &L\left(\{\hat{\boldsymbol{x}}, \hat{\boldsymbol{c}}, \hat{\boldsymbol{d}}, \tilde{\boldsymbol{r}}\}, \{\boldsymbol{x}, \tilde{\boldsymbol{c}}, \tilde{\boldsymbol{d}}\}, \{\boldsymbol{\lambda}, \boldsymbol{\mu}, \boldsymbol{\sigma}\}\right) \\ &= \sum_{m=1}^{M} y_m\left(\hat{x}_{im}, \hat{c}_{im}, \hat{d}_{im}, \tilde{r}_{im}\right) + \sum_{m=1}^{M}\sum_{i=1}^{I} \lambda_{im}\left(\hat{x}_{im} - x_{im}\right) \end{aligned}$$

$$+\sum_{m=1}^{M}\sum_{i=1}^{I}\mu_{im}\left(\hat{c}_{im}-\tilde{c}_{im}\right)+\sum_{m=1}^{M}\sum_{i=1}^{I}\sigma_{im}\left(\hat{d}_{im}-\tilde{d}_{im}\right)$$
$$+\frac{\rho}{2}\sum_{m=1}^{M}\sum_{i=1}^{I}\left\|\hat{x}_{im}-x_{im}\right\|^{2}+\frac{\rho}{2}\sum_{m=1}^{M}\sum_{i=1}^{I}\left\|\hat{c}_{im}-\tilde{c}_{im}\right\|^{2}$$
$$+\frac{\rho}{2}\sum_{m=1}^{M}\sum_{i=1}^{I}\|\hat{d}_{im}-\tilde{d}_{im}\|^{2} \tag{31}$$

where $\boldsymbol{\lambda} = \{\boldsymbol{\lambda_m}\}_{m\in\mathcal{M}} = \{\lambda_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}$, $\boldsymbol{\mu} = \{\boldsymbol{\mu_m}\}_{m\in\mathcal{M}} = \{\mu_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}$, and $\boldsymbol{\sigma} = \{\boldsymbol{\sigma_m}\}_{m\in\mathcal{M}} = \{\sigma_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}$ are the Lagrange multipliers, and ρ is a positive coefficient that can affect the speed of iterative convergence.

For ease of presentation, define local variables set $\boldsymbol{LV}$, global variables set $\boldsymbol{GV}$, and scaled dual variables set $\boldsymbol{SV}$ as follows:

$$\begin{cases}\boldsymbol{LV}=\{\hat{\boldsymbol{x}},\hat{\boldsymbol{c}},\hat{\boldsymbol{d}},\tilde{\boldsymbol{r}}\}\\ \boldsymbol{GV}=\{\boldsymbol{x},\tilde{\boldsymbol{c}},\tilde{\boldsymbol{d}}\}\\ \boldsymbol{SV}=\{\boldsymbol{u},\boldsymbol{v},\boldsymbol{w}\}\end{cases} \tag{32}$$

where $\boldsymbol{u} = \{u_{im}|\ u_{im} = \lambda_{im}/\rho\}_{i\in\mathcal{I},m\in\mathcal{M}}$, $\boldsymbol{v} = \{v_{im}|\ v_{im} = \mu_{im}/\rho\}_{i\in\mathcal{I},m\in\mathcal{M}}$, and $\boldsymbol{w} = \{w_{im}|\ w_{im} = \sigma_{im}/\rho\}_{i\in\mathcal{I},m\in\mathcal{M}}$ are the scaled dual variable, and the new augmented Lagrangian can be rewritten as follows by leveraging $\boldsymbol{LV}$, $\boldsymbol{GV}$, and $\boldsymbol{SV}$:

$$L^{\rho}(\boldsymbol{LV},\boldsymbol{GV},\boldsymbol{SV})$$
$$=\sum_{m=1}^{M}\Bigg[y_m\left(\hat{x}_{im},\hat{c}_{im},\hat{d}_{im},\tilde{r}_{im}\right)+\frac{\rho}{2}\sum_{i=1}^{I}\|\hat{x}_{im}-x_{im}+u_{im}\|^{2}$$
$$+\frac{\rho}{2}\sum_{i=1}^{I}\|\hat{c}_{im}-c_{im}+v_{im}\|^{2}+\frac{\rho}{2}\sum_{i=1}^{I}\|\hat{d}_{im}-d_{im}+w_{im}\|^{2}\Bigg]. \tag{33}$$

Now, we have obtained the augmented Lagrangian function for the optimization problem P_5. Then, based on the above expression, our final goal can be described as the max-min problem of the augmented Lagrangian function

$$\boldsymbol{P_6}:\ \max_{\boldsymbol{SV}}\left[\min_{\boldsymbol{LV},\boldsymbol{GV}} L^{\rho}(\boldsymbol{LV},\boldsymbol{GV},\boldsymbol{SV})\right]. \tag{34}$$

Following the preceding design rationale, our optimization problem is approximatively transformed from P_1 to P_6, we will focus on solving P_6 by iterating multiple variables in what follows, including local variable $\boldsymbol{LV}$, global variable $\boldsymbol{GV}$, and scaled dual variable $\boldsymbol{SV}$. While the iterative process is shown in Fig. 4, including three types of optimization problems, that is, optimizing $\min_{LV}\ L^{\rho}(\boldsymbol{LV},\boldsymbol{GV}^{(t)},\boldsymbol{SV}^{(t)})$ when $\boldsymbol{GV}^{(t)}$ and $\boldsymbol{SV}^{(t)}$ are given, optimizing $\min_{GV}\ L^{\rho}(\boldsymbol{LV}^{(t)},\boldsymbol{GV},\boldsymbol{SV}^{(t)})$ when $\boldsymbol{LV}^{(t)}$ and $\boldsymbol{SV}^{(t)}$ are given, optimizing $\min_{SV}\ L^{\rho}(\boldsymbol{LV}^{(t)},\boldsymbol{GV}^{(t)},\boldsymbol{SV})$ when $\boldsymbol{LV}^{(t)}$ and $\boldsymbol{GV}^{(t)}$ are given. It is worth mentioning that the first optimization problem can be decomposed by multiple satellites based on the following (35), and each satellite m can optimize and make decision by solving $L_m^{\rho}(\boldsymbol{LV}_m,\boldsymbol{GV},\boldsymbol{SV})$ independently:

$$L^{\rho}\left(\boldsymbol{LV},\boldsymbol{GV}^{(t)},\boldsymbol{SV}^{(t)}\right)=\sum_{m=1}^{M}L_m^{\rho}\left(\boldsymbol{LV}_m,\boldsymbol{GV}^{(t)},\boldsymbol{SV}^{(t)}\right). \tag{35}$$

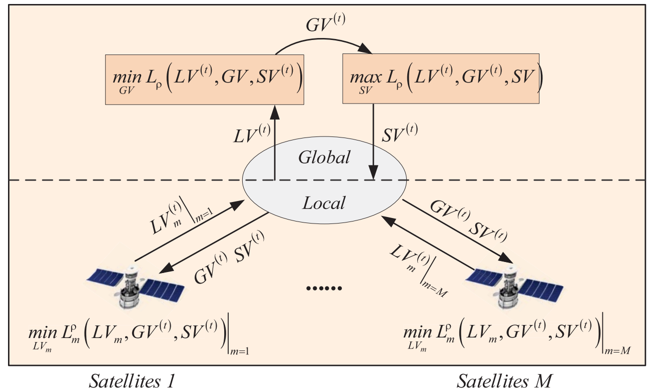

Fig. 4. This figure illustrates the distributed algorithm iterative interaction process. For concision, the satellites responsible for updating the global variables are called distributed orchestrators, while other satellites responsible for optimizing the local variables are called worker satellites.

The detailed variable iteration algorithm is given in what follows, including local variables $\boldsymbol{LV}$ update, global variables $\boldsymbol{GV}$ update, and scaled dual variables $\boldsymbol{SV}$ update.

B. Local Variables Update

The local variables set $\boldsymbol{LV}$ can be divided into M parts, that is, $\boldsymbol{LV} = \{\boldsymbol{LV}_m \mid \boldsymbol{LV}_m = \{\hat{x}_{im},\hat{c}_{im},\hat{d}_{im},\tilde{r}_{im}\}_{i\in\mathcal{I}}\}_{m\in\mathcal{M}}$, where $\boldsymbol{LV}_m$ means the local variables that optimized by satellite m. During the local variables update, we need to solve the following optimization problem and update local variables $\boldsymbol{LV}_m$ given $\boldsymbol{GV}^{(t)}$ and $\boldsymbol{SV}^{(t)}$ for each satellite m:

$$\boldsymbol{LV}_m^{(t+1)}=\operatorname*{argmin}_{LV_m}\ L_m^{\rho}\left(\boldsymbol{LV}_m,\boldsymbol{GV}^{(t)},\boldsymbol{SV}^{(t)}\right). \tag{36}$$

The above optimization can be denoted as the following optimization subproblem P_7:

$$\boldsymbol{P_7}:\min_{\substack{\hat{x}_{im},\hat{c}_{im}\\ \hat{d}_{im},\tilde{r}_{im}}}\Bigg[y_m\left(\hat{x}_{im},\hat{c}_{im},\hat{d}_{im},\tilde{r}_{im}\right)$$
$$+\frac{\rho}{2}\sum_{i=1}^{I}\|\hat{x}_{im}-x_{im}^{(t)}+u_{im}^{(t)}\|^{2}+\frac{\rho}{2}\sum_{i=1}^{I}\|\hat{c}_{im}-\tilde{c}_{im}^{(t)}+v_{im}^{(t)}\|^{2}$$
$$+\frac{\rho}{2}\sum_{i=1}^{I}\|\hat{d}_{im}-\tilde{d}_{im}^{(t)}+w_{im}^{(t)}\|^{2}\Bigg] \tag{37}$$
$$\text{s.t. } \boldsymbol{LV}_m\in\phi_m \tag{38}$$

where $\boldsymbol{LV}_m \in \phi_m$ is the feasible set for local variables mentioned in Section IV-C. Obviously, P_7 is a convex optimization problem, including a convex function with respect to $\boldsymbol{LV}_m$ and a convex feasible set ϕ_m, which can be solved by each satellite based on the convex theory independently. In our paper, the optimization subproblem P_7 is solved according to the interior-point method.

C. Global Variables Update

In this part, we will update the global variables $\boldsymbol{GV}^{(t+1)}$ by $\boldsymbol{LV}^{(t)}$ and $\boldsymbol{SV}^{(t)}$, where $\boldsymbol{GV} = \{\boldsymbol{gv}_{im}|\boldsymbol{gv}_{im} = \{x_{im},\tilde{c}_{im},\tilde{d}_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}\}$

$$\boldsymbol{GV}^{(t+1)}=\operatorname*{argmin}_{GV}\ L^{\rho}\left(\boldsymbol{LV}^{(t)},\boldsymbol{GV},\boldsymbol{SV}^{(t)}\right). \tag{39}$$

Considering that $y_m(\hat{x}_{im}, \hat{c}_{im}, \hat{d}_{im}, \tilde{r}_{im})$ is immaterial to the global variables, (39) can be transformed to the follows three optimization problems when updating $\boldsymbol{gv}_{im} = \{x_{im}, \tilde{c}_{im}, \tilde{d}_{im}\}$, respectively:

$$x_{im}{}^{(t+1)} = \arg\min_{x_{im}} \left[\frac{\rho}{2} \sum_{m=1}^{M} \sum_{i=1}^{I} \left(\hat{x}_{im}^{(t)} - x_{im} + u_{im}^{(t)} \right)^2 \right]$$
$$\text{s.t.} \ \sum_{m=1}^{M} x_{im} \leq 1 \quad \forall i \tag{40}$$

where the above constraints are proposed in (26) for all users i, which is used to limit the offloading between different satellites according to global variables. Since there is a summation of variable i in the objective function, and there is $\forall i$ in the constraints, then we can directly decompose the above problems according to variable i and avoid the high complexity of above optimization problem. In addition, we can update the global variables of $\tilde{c}_{im}, \tilde{d}_{im}$ according to the following optimization problem:

$$\tilde{c}_{im}^{(t+1)} = \arg\min_{\tilde{c}_{im}} \times \left[\frac{\rho}{2} \sum_{m=1}^{M} \sum_{i=1}^{I} \left(\hat{c}_{im}^{(t)} - \tilde{c}_{im} + v_{im}^{(t)} \right)^2 \right] \tag{41}$$

$$\tilde{d}_{im}^{(t+1)} = \arg\min_{\tilde{d}_{im}} \times \left[\frac{\rho}{2} \sum_{m=1}^{M} \sum_{i=1}^{I} \left(\hat{d}_{im}^{(t)} - \tilde{d}_{im} + w_{im}^{(t)} \right)^2 \right]. \tag{42}$$

It is worth mentioning that the above formulas running on the global satellite need to collect the results of local variables sending from multiple satellites. Take the partial derivative of (41) and (42) with respect to $\tilde{c}_{im}$ and $\tilde{d}_{im}$, respectively, and set them equal to zeros, then we can derive the analytical solution that can be used to update the global variables of $\tilde{c}_{im}$ and $\tilde{d}_{im}$

$$\tilde{c}_{in}^{(t+1)} = \hat{c}_{in}^{m(t+1)} + v_{in}^{m(t)} \quad \forall n, i \tag{43a}$$
$$\tilde{d}_{in}^{(t+1)} = \hat{d}_{in}^{m(t+1)} + w_{in}^{m(t)} \quad \forall n, i. \tag{43b}$$

It is worth mentioning that distributed orchestrators use (43a) and (43b) and the solution of (40) to update the global variables and transmit them to other worker satellites.

D. Scaled Dual Variables Update and Stopping Criterion

In this section, we will update the scaled dual variables $\boldsymbol{SV} = \{\boldsymbol{sv}_m \mid \boldsymbol{sv}_m = \{u_{im}, v_{im}, w_{im}\}_{i\in\mathcal{I}}\}_{m\in\mathcal{M}}$ as follows:

$$u_{im}^{(t+1)} = u_{im}^{(t)} + \rho\left(\hat{x}_{im} - x_{im}\right) \quad \forall m, i \tag{44a}$$
$$v_{im}^{(t+1)} = v_{im}^{(t)} + \rho\left(\hat{c}_{im} - \tilde{c}_{im}\right) \quad \forall m, i \tag{44b}$$
$$w_{im}^{(t+1)} = w_{im}^{(t)} + \rho\left(\hat{d}_{im} - \tilde{d}_{im}\right) \quad \forall m, i. \tag{44c}$$

It is worth mentioning that distributed orchestrators use (44a)–(44c) update Lagrange multipliers and transmit them to other worker satellites, and then worker satellites use them update local variables, independently.

Finally, the stopping criteria, including local criterion and global criterion are introduced as follows, where the local criterion is used for local variables for each satellite, denoted as

$$\left\| \widehat{\boldsymbol{x}}_{\boldsymbol{m}}^{(\boldsymbol{t+1})} - \boldsymbol{x}^{(\boldsymbol{t+1})} \right\|^2 \leq \epsilon_{\text{local}} \quad \forall m \tag{45a}$$

Algorithm 1 Binary Variables Recovery

Input: $\boldsymbol{x} = \{x_{im}\}_{i\in\mathcal{I},m\in\mathcal{M}}$
Output: $\boldsymbol{x}^{recover}$
Initialize $\boldsymbol{x}^{recover} = \boldsymbol{0}$
for i = 1, 2,..., I **do**
  Find the largest $\{x_{im}\}_{m\in\mathcal{M}}$, and set it to 1.
end for
Compute the derivations of the original objective function $D_{im}^x = \partial T_1^{total}(x) \ /\partial x_{im}$ with respect to each x_{im}.
Considering that our problem is to minimize the total delay and D_{im}^x is not positive, thus we sort all D_{im}^x from the smallest to largest, and record them as $\{D_1^x, D_2^x, \ldots, D_I^x\}$. The corresponding sorted x_{im} is recorded as $x_1', x_2', \ldots, x_I'$.
for i = 1, 2,..., I **do**
  Set $x_i' = 1$ ($x_i' \in \boldsymbol{x}^{recover}$) if the constrains in formula (10) hold expect for the forth constrain due to the approximate transformation in Section IV for $h(\hat{x}_{im}, \tilde{r}_{im})$. Otherwise, continue to the next loop.
end for
Scale r_{im} to satisfy the forth constrain in formula (10), that is, set $r_{im}' = \frac{r_{im}}{\sum_i r_{im}}$ for each m and i as the final solution of problem P_1 about transmission rate.
return $\boldsymbol{x}^{recover}$

$$\left\| \widehat{\boldsymbol{c}}_{\boldsymbol{m}}^{(\boldsymbol{t+1})} - \boldsymbol{c}^{(\boldsymbol{t+1})} \right\|^2 \leq \epsilon_{\text{local}} \quad \forall m \tag{45b}$$
$$\left\| \widehat{\boldsymbol{d}}_{\boldsymbol{m}}^{(\boldsymbol{t+1})} - \boldsymbol{d}^{(\boldsymbol{t+1})} \right\|^2 \leq \epsilon_{\text{local}} \quad \forall m \tag{45c}$$

where ϵ_{global} as a tolerance for the local optimization problem is a small positive constant. Besides, the global criterion is used for global variables for each distributed orchestrator, denoted as

$$\left\| \boldsymbol{x}^{(\boldsymbol{t+1})} - \boldsymbol{x}^{(\boldsymbol{t})} \right\|^2 \leq \epsilon_{\text{global}} \tag{46a}$$
$$\left\| \boldsymbol{c}^{(\boldsymbol{t+1})} - \boldsymbol{c}^{(\boldsymbol{t})} \right\|^2 \leq \epsilon_{\text{global}} \tag{46b}$$
$$\left\| \boldsymbol{d}^{(\boldsymbol{t+1})} - \boldsymbol{d}^{(\boldsymbol{t})} \right\|^2 \leq \epsilon_{\text{global}} \tag{46c}$$

where ϵ_{local} as a tolerance for the global optimization problem is a small positive constant.

E. Binary Variables Recovery and Complexity Analysis

Considering that we have relaxed the bivariate x_{im} in order to obtain a convex optimization problem and solve the mixed integer programming approximatively in Section IV-B. Now, let's give the recover algorithm for variables x_{im} and d_{im} to recover the original optimization problem. Here, we give variables x_{im} as an example, and the same method can be used to d_{im}. We can see that the performance gap between the relaxation method and the original optimization problem are not obvious in the simulation results.

Now, we discuss the complexity analysis of our distributed optimization algorithm, similar to the analysis found in [10]. Considering that there are I users and M satellites for each ADMM-based optimized unit, the complexity of centralized optimization is $\mathcal{O}(I^3M^3)$, whereas the complexity of the

ADMM-based optimization algorithm is $k\mathcal{O}(I^3)$, where k stands for the number of iterations. The distributed ADMM algorithm clearly reduces the computational complexity of the original problem.

VI. Simulation Results and Analysis

In this section, we compare our proposed approach with three benchmark schemes: a centralized method based on the interior-point method, a ground processing algorithm that completely disregards offloading, and the distributed method in [10], which does not jointly consider communication, computation, and storage constraints and only supports satellites providing services to users within their coverage area (assuming no intersatellite communications). We verify the performance of our proposed scheme using the average task completion delay as the standard metric. All the simulations were conducted 100 times on a server equipped with an Intel i7-12700K CPU, 128 GB of RAM, and an RTX 4080S GPU. In addition, the main parameter settings are as follows.

1) *Satellites Parameters:* We consider a satellite network scenario consisting of 20 orbital planes, each with 18 low Earth orbit satellites at an altitude of 1000 km. We assume that the satellite network achieves seamless global coverage at this orbital altitude. Each satellite has a computational capacity of ten Gcycles and can support the deployment of up to three task images. The parameter k_2 for the images caching delay is set to 0.06.
2) *Users Parameters:* The transmission power of each user i is 23 dBm, the bandwidth B is 30 MHz, and the large scale fading is considered between satellites and ground users. In addition, the computation capability of each user draws from a discrete integer uniform distribution of $[0.4, 0.8]$ Gcycles.
3) *Task Parameters:* We assume that the data size of service images I_i draws from a discrete integer uniform distribution on $[30, 50]$ Mbit, the number of CPU cycles ϕ_i requested by user i draws from a discrete uniform distribution on $[3, 5]$ Gcycles, and the real-time data D_i draws from a discrete uniform distribution on $[8, 12]$ Mbit.

First, we simulate the impact of the number of users on the average task completion delay based on four algorithms. User distribution is based on the global population density distribution. The ADMM joint optimization represents the proposed scheme in this article, while the other three are benchmark schemes, as shown in Fig. 5. Clearly, as the number of users increases, the task completion delay shows an upward trend. The green line represents user tasks computed locally, unaffected by satellite resources, and thus remains constant under the given parameter settings. The red and purple lines almost overlap, indicating that our proposed distributed scheme has minimal performance loss compared to the centralized algorithm. The blue line also represents a distributed scheme but has significantly higher task completion delay due to the lack of joint optimization of multidimensional resources and the assumption that intersatellite communication

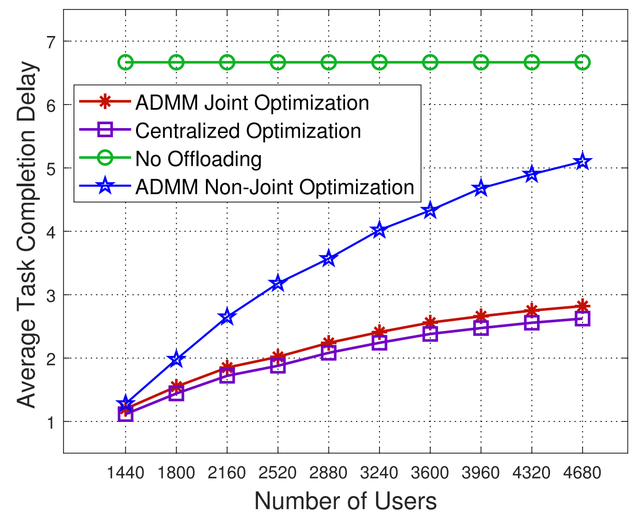

Fig. 5. Average task completion delay versus number of users.

is not possible. It is noteworthy that the growth curves, except for the green line, become progressively flatter. This is because satellite resources are gradually exhausted, and the primary factor affecting task completion delay shifts from satellite resources to local device computing capacity, continually approaching the task completion delay of completely local computation.

To more intuitively illustrate the difference in offloading decisions between the two distributed schemes, we simulate the load situation of the satellite network, as shown in Fig. 6. And We set number of users is 3240. The shades of light blue indicate the satellite load, with darker colors representing heavier loads. We can see that the distributed scheme in literature [10] assumes no communication between satellites, only serving users within their coverage area. This results in extremely unbalanced satellite loads and wastage of idle satellite resources. In contrast, our distributed strategy fully utilizes all satellites in space. Although there is a certain transmission delay cost, the overall satellite load is more balanced, leading to superior system performance as shown in Fig. 5.

Next, we delve into the reasons why the performance of the centralized algorithm is nearly identical to that of our proposed distributed scheme. To facilitate understanding, we use a simple scenario with 20 users and four satellites as an example for illustration. In Fig. 7, we compare the value of global decision variables for centralized optimization and the value of global decision variables for ADMM-based joint optimization, including $\boldsymbol{x}$, $\boldsymbol{c}$, and $\boldsymbol{d}$. We randomly select four ground users and compare the solutions between the above two methods. We can conclude that the solutions obtained by them are almost equal expect for slight discrepancies, especially for the results of computation resource allocation $\boldsymbol{c}$ in Fig. 7(a). In addition, since each user can offload their services to the set of satellite $\mathcal{M}$, but only choose one, which means vector $\{x\}_{m\in\mathcal{M}}$ is a 0-1 integer vector. We can relax the decision variables x in Section IV, which causes vector $\{x\}_{m\in\mathcal{M}}$ becomes nonzero continuous vectors instead of original 0-1 integer vectors for

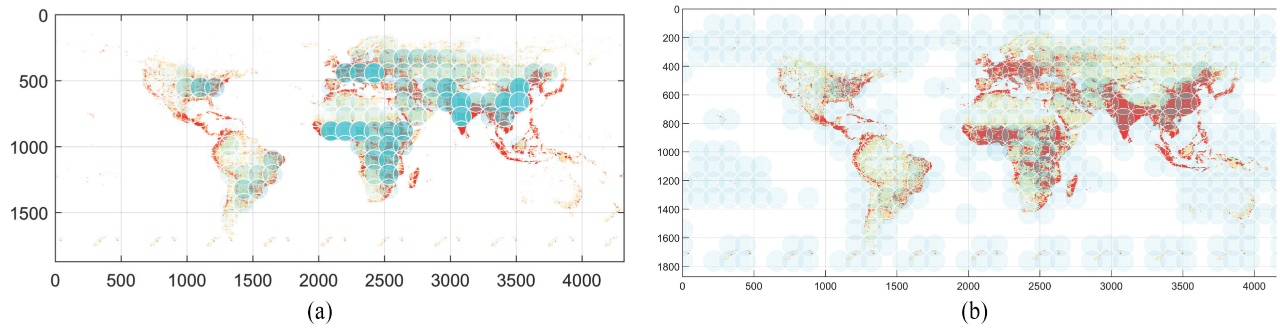

Fig. 6. Performance comparison of satellite network load under task offloading. (a) Satellites load based on the literature [10]. (b) Satellites load based on our algorithm.

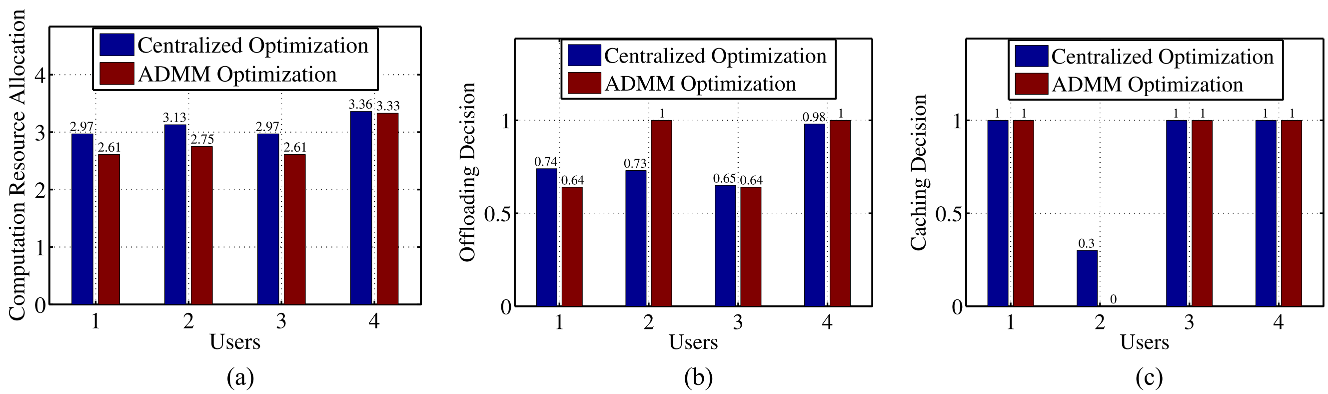

Fig. 7. Optimization results of global decision variables for centralized optimization and ADMM-based joint optimization. (a) Computation resource allocation. (b) Offloading decision. (c) Caching decision.

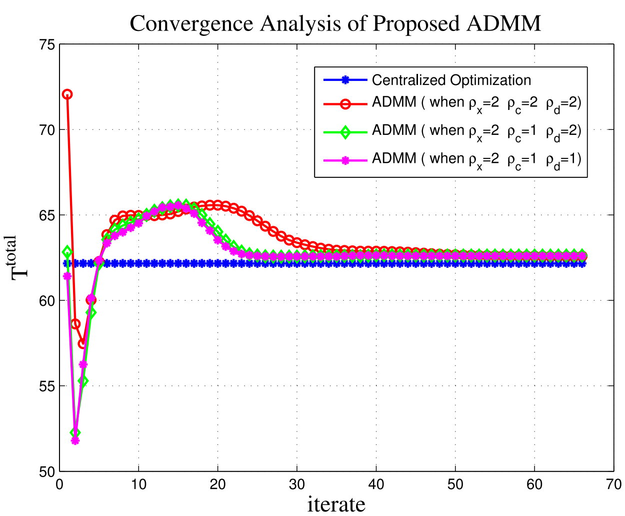

Fig. 8. Convergence of the proposed ADMM method with the value of different parameters.

each user i. Thus, to simplify the illustration, we choose and compare the largest x_{im} in vector $\{x\}_{m\in\mathcal{M}}$ in Fig. 7(b), which means the most possible offloading preference for user i. As we can see that the decision variables $\boldsymbol{x}$ of the ADMM-based joint optimization are very close to that of the centralized optimization. This explains why the results obtained from the two methods are similar.

Then, we verify the convergence performance of our joint optimization ADMM-based algorithm with the different parameters ρ as illustrated in Fig. 8, where the X-coordinate is the number of iterations. We consider a simple scenario, including 20 users and four satellites as an example. As we can see that the blue line as the optimal solution is the performance of centralized optimization, which is not related to the number of iterations. While other three curves can converge to the optimal solution with different convergence rates with respect to the number of iterations. Since the value of x, c, and d are in different orders of magnitude, we set different ρ corresponding as ρ_x, ρ_c, and ρ_d for (31), respectively. Considering that the value of c is the largest and the value of x is the smallest, thus we set ρ_c as the smallest value and ρ_x as the largest value to balance the augmented lagrangian function. If the value of ρ_x, ρ_c, and ρ_d are same, then the joint optimization ADMM-based optimization will not converge to the optimal solution with a poor performance.

Finally, we investigated the impact of variations in communication capacity and satellite computing capacity on the system performance, while also demonstrating the advantages of our scheme, as shown in Fig. 9. For the convenience of comparing the results under different schemes, we set the number of users to 3240 and the local computing capacity to 0.8. Fig. 9 shows that as either the communication capacity or the computing capacity increases, the average task completion delay consistently tends to decrease. The green curve represents tasks processed locally without offloading, thus

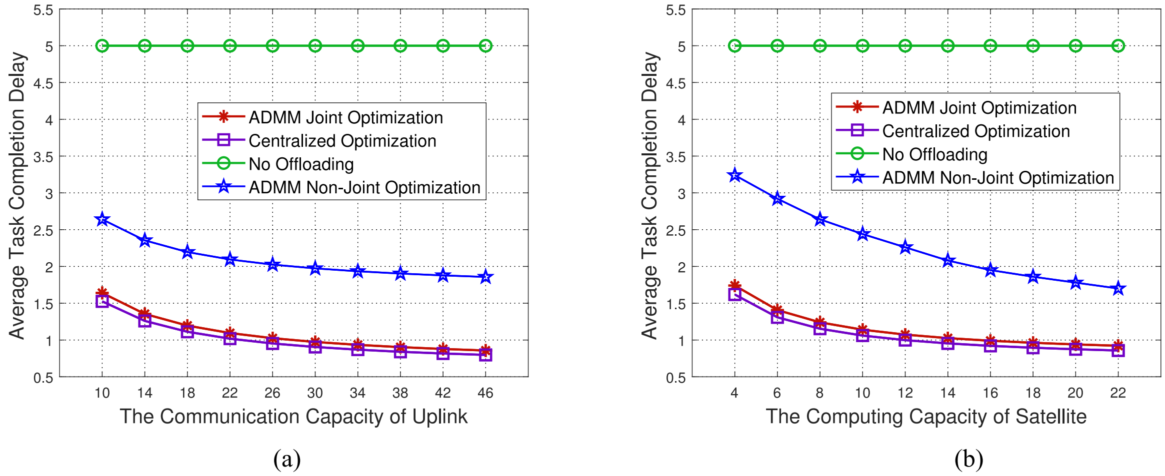

Fig. 9. Impact of parameters on average task completion delay. (a) Uplink communication capacity. (b) Satellite computing capacity.

remaining unchanged with varying parameters. As previously analyzed, the red and purple curves almost overlap, indicating that the results of our distributed joint optimization algorithm are close to those of the centralized optimization. The blue curve, representing the distributed scheme, shows a task completion delay that is 60%–90% greater than that of our proposed scheme, demonstrating inferior performance. All curves, except for the green curve, experience a rapid initial decline followed by a slower decrease. This is because the number of users (tasks) is fixed; initially, the increase in resources leads to a rapid decrease in delay, but as resources continue to increase, the excess resources cannot be fully utilized by the tasks, causing the delay to approach a stable lower bound. Additionally, an interesting phenomenon is observed: the increase in the communication capacity [Fig. 9(a)] does not widen the gap between the blue and red lines. In contrast, as the satellite computing capacity increases [Fig. 9(b)], the gap between the two lines tends to narrow. This is because the ADMM nonjoint optimization algorithm only supports users within the coverage area of satellites, and even with continuous enhancement of communication capacity, the number of users that satellites can serve has an upper limit. When computing capacity significantly increases, users' tasks do not need to be transferred through intersatellite links to satellites outside the coverage area, making our proposed algorithm nearly equivalent to the blue line strategy, thus reducing the gap between the two.

VII. Conclusion

In this article, we focus on service deployment to edge satellites for terrestrial users. To decrease the total network delay for terrestrial users, a joint nonconvex optimization model, including service deployment decision, computation resource allocation, transmission rate allocation, as well as content caching is proposed. Then, we transform it to a convex optimization problem by variable substitution and binary variable relaxation. Next, a distributed optimization model based on ADMM is proposed, which aims at relieving the computing burden of single satellite by leveraging multisatellites collaboratively. Finally, the simulation results verify the efficiency of our distributed model comparing with traditional baseline models.

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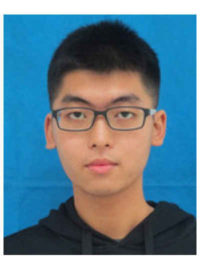

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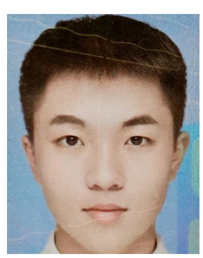

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